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The Open Network on Data Exchange.

Turning multi-agency data into trusted, coordinated public service.

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What we will cover

01 The problem

02 The solution

03 The tool: 1864 Prep

04 The network

05 Partners and timing

06 Evaluation

07 Next steps

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01

THE PROBLEM

Agencies cannot safely check each other’s records. Data sits idle, privacy is exposed, and people pay.

When agencies can’t verify each other, everyone pays.

THE COST TO GOVERNMENT

No trusted way to check
No agency can confirm another’s record is genuine.

Privacy exposed
Agencies copy whole databases just to check one fact.

Data that cannot match
Spreadsheets and paper records that do not line up.

Programmes stall
Targeting fails, budgets overspend, pilots are abandoned.

Leakage
Up to a third of subsidy lost to duplication and resale.

THE COST TO PEOPLE

Sent back and forth (the loop)
People re-prove what the state already knows.

Wrongly excluded
A genuine farmer with no matching record is turned away.

Served twice
One household collects fertilizer under several names.

Identity misused
Fertilizer collected in a relative’s or stranger’s name.

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02

THE SOLUTION

Fix the data at each agency, then let agencies trust each other’s answers without sharing data.

First standardise, then connect.



PART 1 · THE TOOL

1864 Prep

Cleans and standardises each agency's data, on its own computer.

Messy files in; match-ready data out.





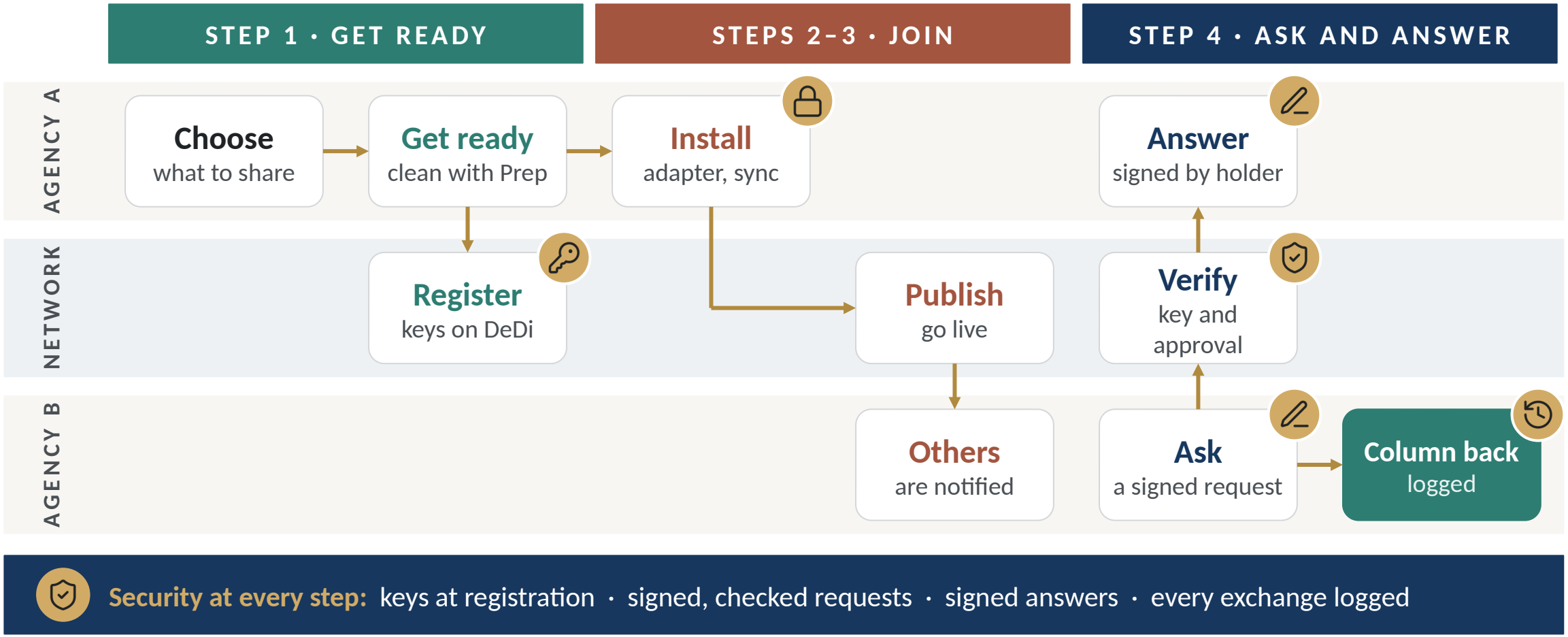
PART 2 · THE NETWORK

The Open Network

Carries a trusted, verified answer between agencies, with no central database.

A question in; a signed answer out.

How it works: an overview.



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THE TOOL: 1864 PREP

Messy, low-capacity data is the silent killer of every digital project. This is where we fix it.

STEP 1 OF 5 GET READY · LIVE

✔ WORKING TODAY

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Get the data ready

Why it matters

A network is only as good as the data behind it. Many agencies still run on spreadsheets or paper, so even their own records are hard to use.

If an ID loses a digit or a place is misspelt, the match fails and a real person is missed. 1864 Prep fixes this on the agency's own machine.

▶ WE WILL SHOW THIS LIVE

```
// a real file, before → after

Kastina   → Katsina
071       → 0071 (ID kept)
musa a.   → Musa Abubakar
12/3/24   → 2024-03-12

✓ 14,200 records standardised
✓ 58 duplicates flagged
```

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The machine learning inside 1864 Prep.

UNDERSTANDS

What each field means

A trained classifier tells a date, name, amount or code.

FINDS

Relevant facts in free text

People, places, organisations and money inside text.

READS

Most files, even broken headers

CSV, Excel, JSON and PDF tables, in any encoding.

FIXES

Mismatched but identical labels

Similar labels grouped for staff to confirm.

MATCHES

Irregular values to official lists

States, LGAs and currencies, matched to the register.

REFUSES

To guess

Problems are flagged for review; IDs are never changed.

1864 Prep turns any messy file into network-ready data.

1 · TAKES IN

Any CSV, Excel, JSON or PDF. The table engine finds the real table and skips banners and pivot tabs.

2 · CLEANS

Fixes encodings and dates and keeps long IDs intact. Profiling shows raw and cleaned side by side.

3 · ALIGNS

Matches values to official registers and flags duplicates.

4 · APPROVES

Staff approve every change. The decision ledger makes every change reversible.

TESTED AND OPEN SOURCE

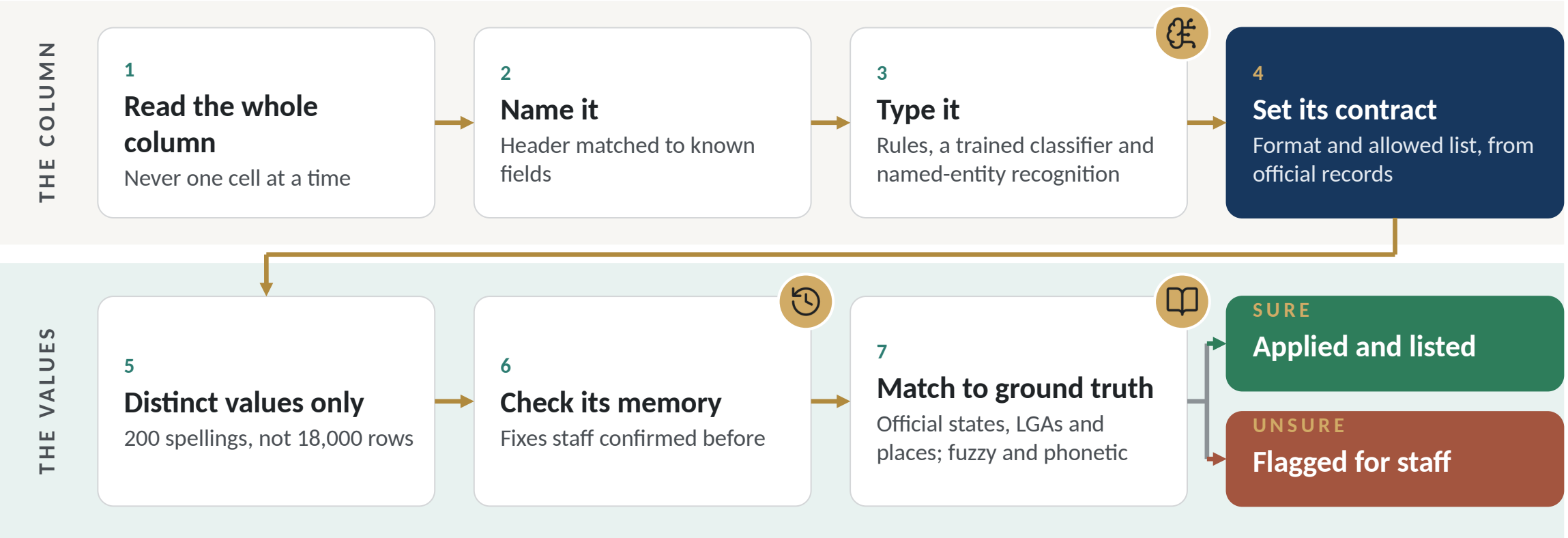
96%

average accuracy, scored on every code change

12 capability areas

85 automated tests

How does 1864 Prep work out what each value should be?



No row-by-row guessing. Context decides each column; ground truth decides each value; staff confirm, and the memory learns.

Four more capabilities under the hood.

AI ASSISTANT

For the hardest columns

Privacy-masked queries, and entirely optional.

TOOL SUITE

A standalone data toolkit

Ten one-click tools, ready-made.

STREAMING

Built for huge files

Reads in batches, and never crashes.

GUIDED FLOWS

A guided path for every task

Step by step, never one black box.

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LIVE DEMO



NOW, LET ME SHOW YOU

1864 Prep

Messy government data is the silent killer of every digital project. Watch it fixed in seconds.

▶️ one864prep-demo.onrender.com

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04

THE NETWORK

The hard question: how do agencies answer each other without ever handing over their data?

What do we build on, and what do we add?

THE AGENCY INSTALLS

One small connector

It changes nothing else in the agency.

EACH AGENCY

1864 ADDS

The trust layer

Every message signed; every exchange logged.

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ALREADY EXISTS

Beckn, the open rail

The open protocol behind ONDC, India's network for digital commerce.

OPEN STANDARD

How does an agency know who sent an answer?

THE PROBLEM	WHAT DEDI DOES
A single database to trust, hack or control	**DECENTRALISATION** Organisations publish on their own domains, not one central database.
How do I know this answer is genuine?	**CRYPTOGRAPHIC VERIFICATION** Digital signatures prove it came from its publisher, unaltered.
Different systems cannot understand each other	**INTEROPERABILITY** Everyone reads and writes through one common protocol and API.

How does an agency join the directory?



MOVE 1

Verify your domain

A DNS record proves the agency owns its web address.



MOVE 2

Generate your keys

The private key stays with the agency; the public key is published.



MOVE 3

Get a record ID

So any agency on the network can find you and trust you.

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✓ FIVE AGENCIES REGISTERED

STEP 2 OF 5 · PROVE WHO YOU ARE

What did it take to put agencies on DeDi?

1
Set up the tools
Docker and Beckn ONIX

2
Deploy the adapter
ONIX in containers

3
Open the receiver
The /bpp/receiver endpoint

4
Set the rules
Routing, signing, validation

5
Generate keys
A signing key per participant

6
Go online
A secure ngrok tunnel

7
Connect to DeDi
Identities and keys linked

8
Register agencies
Five participants live

Agriculture (KADA)

State identity (KADRIMA)

Health insurance (NHIA)

National identity (NIMC)

Population (NPC)

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LIVE DEMO



NOW, LET ME SHOW YOU

The directory

The hard part of trust, solved. Watch an agency prove who it is, with no central gatekeeper.

▶️ publish.dedi.global

How does an agency connect, without its data leaving?



1 · THEIR SERVER

Agencies with an API

- Installs on their server
- Calls their live system
- Always current



2 · THEIR COMPUTER

Agencies on Excel

- Installs on an office PC
- Reads the file on site
- Most agencies start here



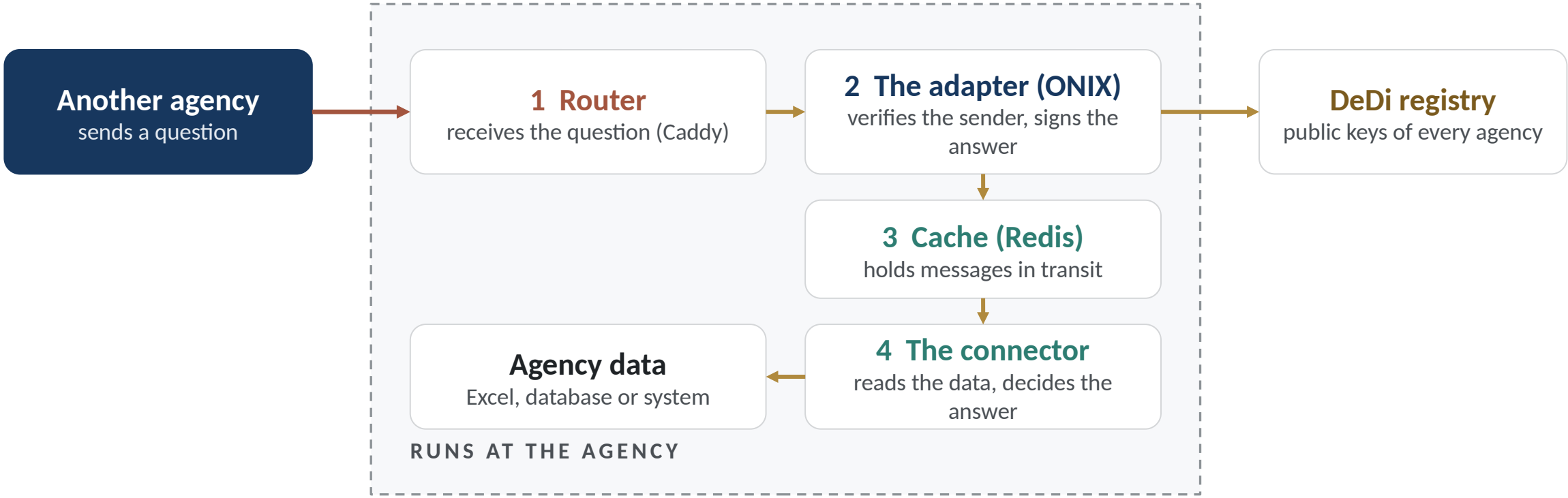
3 · A MANAGED NODE

Non-sensitive registers

- 1864 runs the node
- Reads a synced copy
- Hosted by 1864

In every case, 1864 routes and verifies. It never sees or stores the agency's data.

What happens when a question arrives?



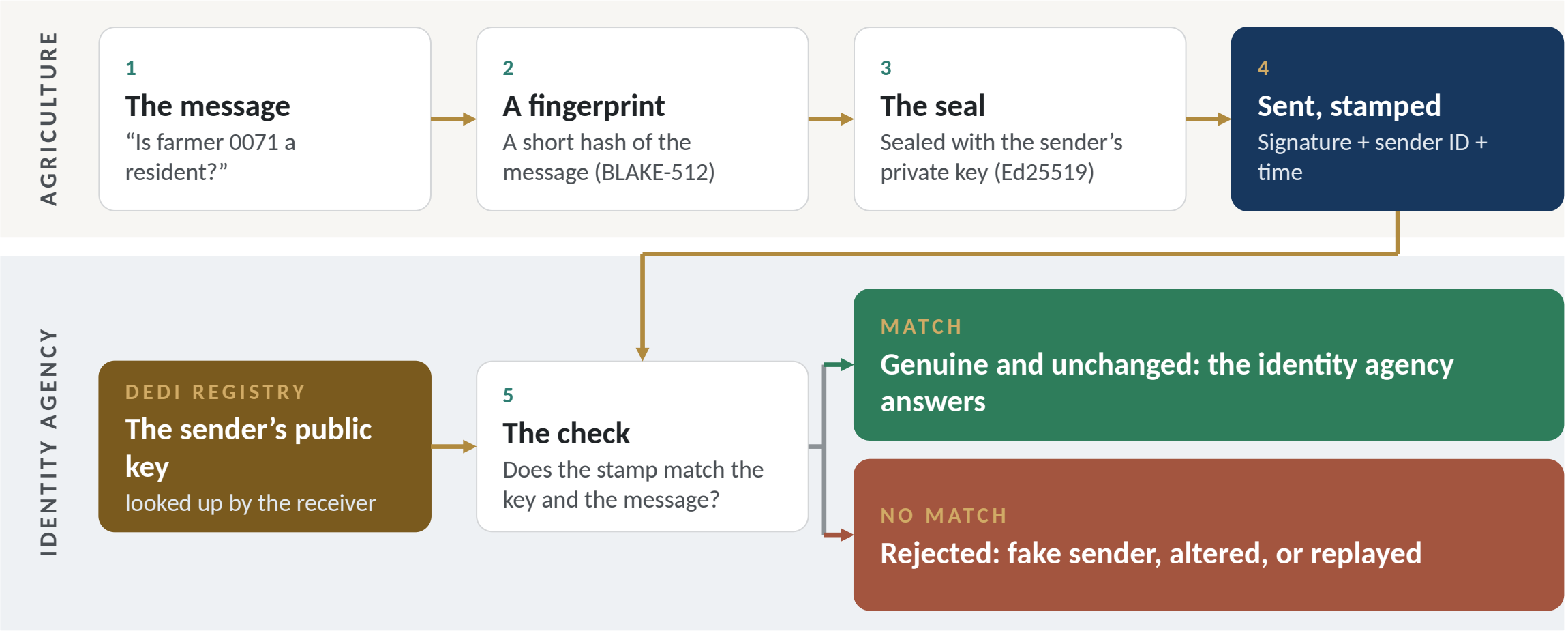
Only the answer leaves. Router, adapter, cache and connector run as one small bundle beside the data. Now: containers on one machine. Next: one packaged install.

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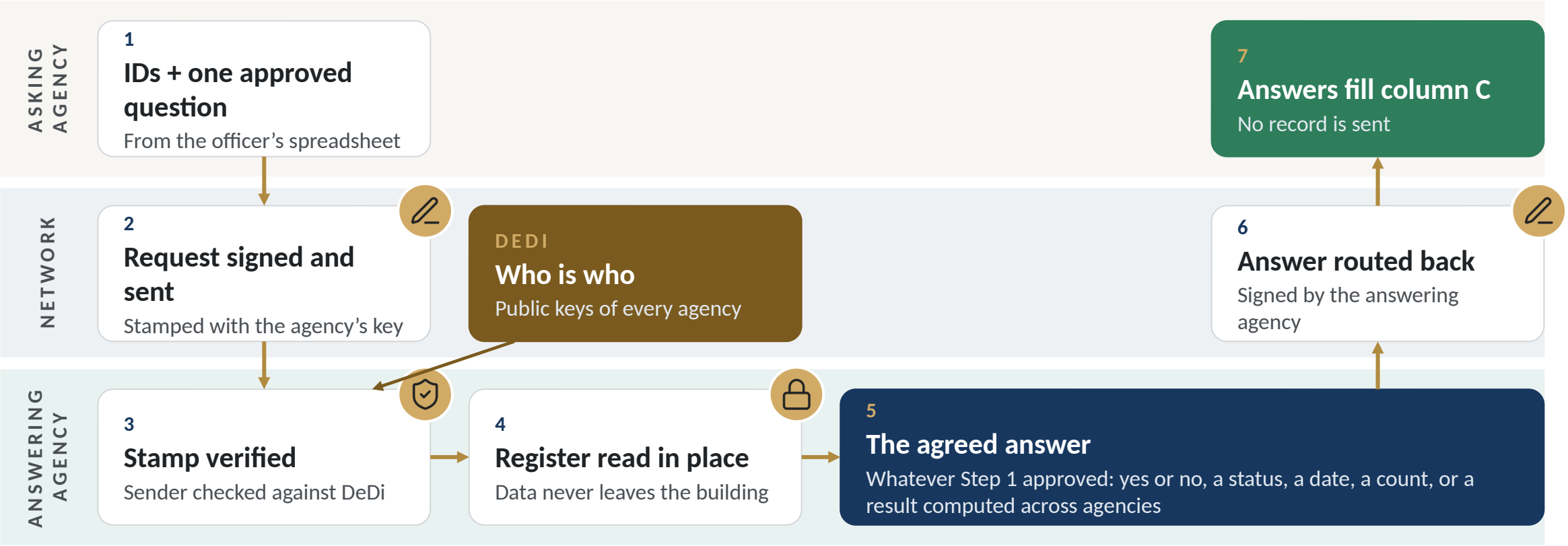
✓ PROVEN IN PROTOTYPE

STEP 4 OF 5 · ASK AND ANSWER

How does a digital stamp prove who sent a message?



What happens when an officer clicks “Ask the network”?



The Open Network: shared rules (Beckn), a registry of who is who (DeDi) and an adapter at each agency. No central database.

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WALKTHROUGH



LET ME WALK YOU THROUGH

From Excel to verified answers

Even an office with nothing but a spreadsheet can answer a verified question. Here is how.

▶️ excel-walkthroug.onrender.com

How is privacy protected?

BUILT IN

Beckn and ONIX

Secure messages

- Digitally stamped by the sender; any change breaks it
- Stamp checked against DeDi
- Encrypted: unreadable on the way

OUR LAYER

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Answers, not files

- Approved questions only
- Only the agreed answer; never the record
- Every exchange logged

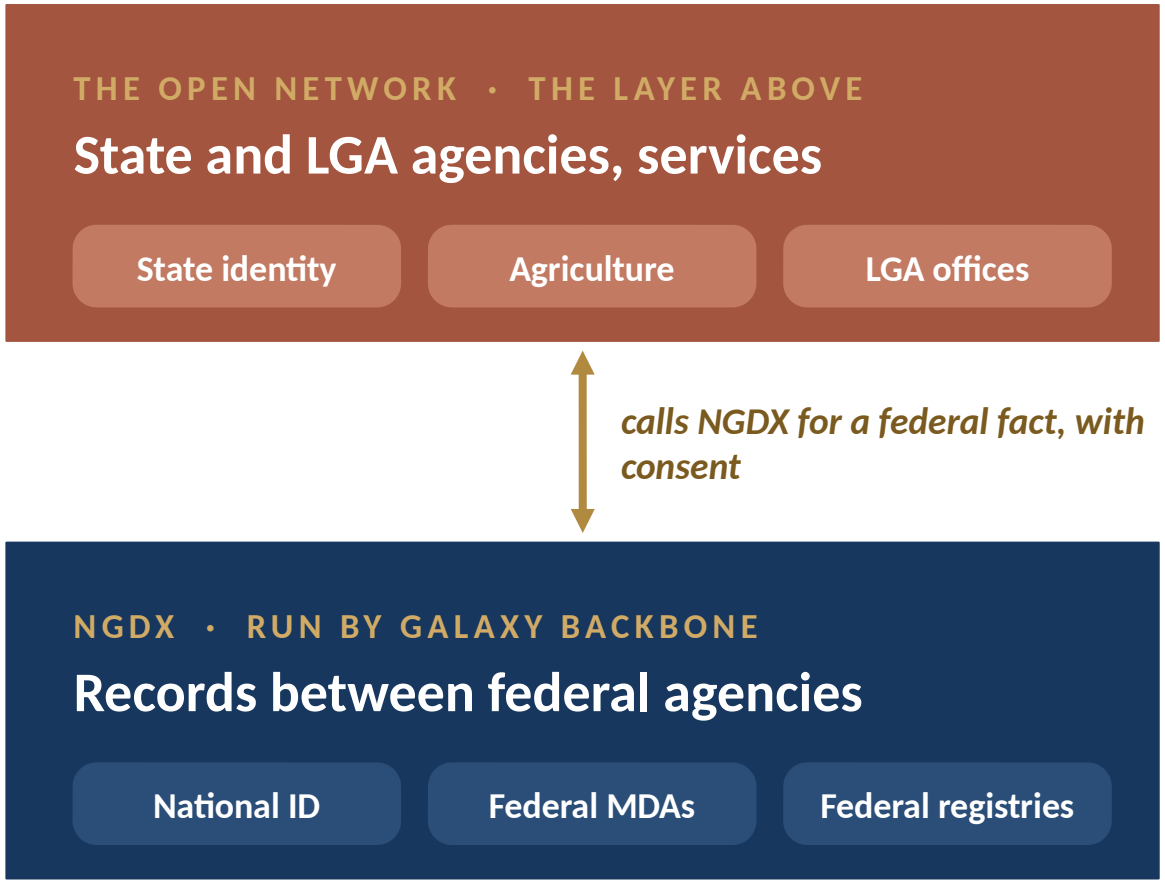
TECHNICAL PARTNER

Center for Effective Global Action (CEGA), UC Berkeley

Privacy technologies

- Compare records without seeing them (multi-party computation)
- Statistics that identify no one (differential privacy)

How does it fit with NGDX, the national rail?



1

It complements NGDX

NGDX connects federal agencies. We bring in the state and LGA offices it does not yet reach.

2

It connects when needed

When a state check needs a federal fact, such as a national ID, it asks NGDX, with consent.

3

It works today

It runs with or without NGDX, so Kaduna does not have to wait.

4

It is ready for NGDX

When government chooses, the network moves onto NGDX.

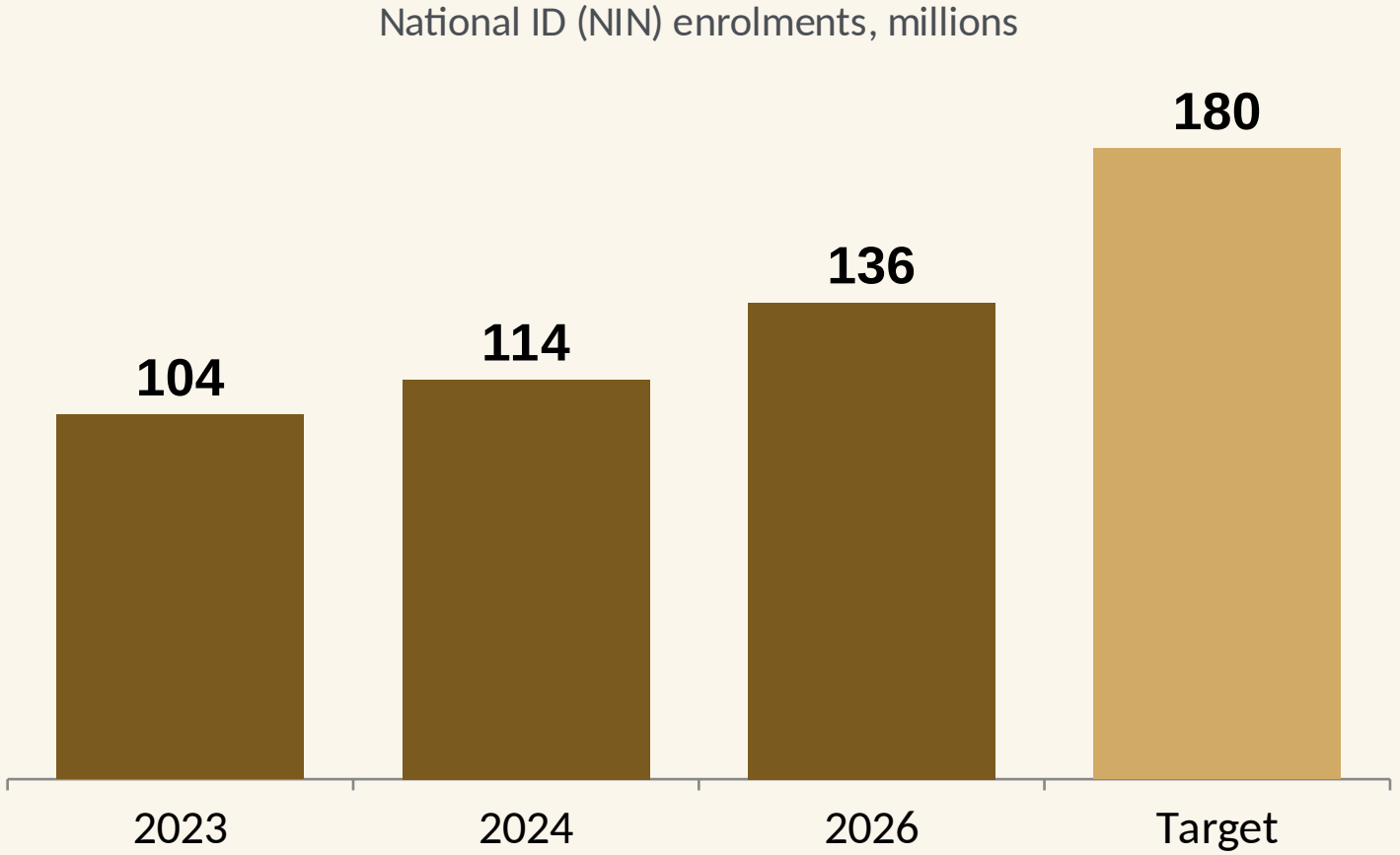
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05

PARTNERS AND TIMING

Why this is the time to build, and who we build it with.

The moment is now: enrolment done, exchange mandated.



NEW IDENTITY LAW, 2026

Exchange is now law

TECHNICAL PARTNER

CEGA, UC Berkeley

Privacy and evaluation.

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EVALUATION

We measure exactly how many more of the right people it reaches.

Can identity verification improve fertilizer targeting?

STAGE 1

Records linked

Farmer list matched to the state identity register.

STAGE 2

A verified list

Duplicates and invalid IDs removed.

STAGE 3

Better targeting

Less leakage; more eligible farmers reached.

STAGE 4

Better farming

Fertilizer use, yields, income, food security.

A randomised trial across 53 subdivisions measures what a verified list delivers.

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07

NEXT STEPS

Where we go from here, and how you can help.

Next: higher controls and security, as we scale.

01

Agency identity

Verify each mandate; issue and revoke credentials.

02

Who sees what

Field-level rules tied to the purpose of each request.

03

Consent and legal basis

Record and enforce consent, or another lawful ground.

04

Durable audit

A tamper-evident history auditors can verify.

05

Privacy by design

Only the fields needed; limited retention; no copies.

06

Secure and resilient

Monitoring, incident response, recovery, certification.

Already in Beckn and ONIX: signing, discovery, encryption, key vault, policy, audit logs.

Next: the governance that lets it scale.

1

Who runs it

A governing body and a trust authority, as more agencies join.

2

Who may join

Formal accreditation, and clear rules to suspend or remove.

3

Rules of sharing

Data-sharing agreements, one owner per dataset, standards.

4

Who is accountable

Liability, independent oversight, disputes and citizen redress.

5

Who sustains it

A funding model, operational ownership and trained teams.

Beckn gives us the rail. Funding builds the trust and governance around it.

From a working prototype to measurable proof.

1

Co-fund the Kaduna pilot

Alongside other funders, from the first signed MoU to measurable results.

2

Back our convening

We bring government to the table. Your name opens doors.

3

Shape the design

Now, from the start.

THE TEAM

Abiola Oyebanjo
Executive Director: strategy and government

Nneka Osadolor
Director of Research: design and methodology

Akintoye Oyekunle
Director of Operations: delivery

Where we are, and what comes next.

✔ **ALREADY UNDER WAY**

The whole loop runs end to end in our prototype, with no records moved. Our first Kaduna MoU is signed, with the social investment agency (KADSIPA).

WHAT NEEDS FUNDING: THE PILOT

- Agency onboarding and connectors
- 1864 Prep, hardened
- Network operations and uptime
- Security, consent and audit
- The pilot and its evaluation

1864 AS NETWORK OPERATOR

- Run the registry and routing
- Onboard agencies, build connectors
- Issue and guard agency keys
- Monitor every node
- Keep consent and the audit trail

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THANK YOU

The right people, served faster.
Let’s build it together.

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